Decision Models as Universal Tool Selectors

Aziz Muminov Alexandra Ilina

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Abstract

Agents increasingly work through many tools: a coding agent connected to a company's MCP servers can face hundreds or thousands of tool definitions. Loading them all is expensive: for 613 tools the largest prompt reached about 182k tokens and a run cost 8–9 times as much as with search. Shipped harnesses therefore search instead, mostly with BM25, as in Codex, Anthropic's tool search and Hermes Agent. Lexical search is cheap but imprecise, and every extra search is a full agent turn. We propose decision models, which return a probability per option in one forward pass, as general-purpose tool selectors. Our selector, built on Microsoft Decision 1, chooses the service, then the tools, with a "none" option and usage-based recommendations, and needs about three decision-model calls per search. On the real tool definitions of 30 MCP servers with 1,076 tools it ranks far better than BM25 offline, with the right tool first for 92% of hand-written queries against 49%. Inside a shipped agent, Hermes Agent, on 37 coding tasks with simulated back ends, pre-planned paired tests run at the same time showed 12.6% fewer tokens with DeepSeek V4.1 Flash over 555 pairs, $p = 0.0002$, and 9.7% fewer with GPT-6.1 Sol over 185 pairs, $p = 0.004$, with 36–60% fewer searches and equal pass rates. The selector's fee made runs 20% more expensive with the inexpensive model and left cost unchanged with the frontier model; savings grow with the agent model's price and with how much it searches. These results cover coding agents, simulated services, two agent models and one harness for the main tests.

1 Introduction

Through the Model Context Protocol, MCP [18], an agent can read CI logs, error trackers, tickets, chat, dashboards, clusters and databases. Each server exposes tens to hundreds of tools, and agents connect to many servers, so which tools the agent sees, and how it finds the rest, now shapes cost and behaviour.

There are two options. The first is to load every definition into every request. For the 613 tools of twelve popular servers the largest prompt then reaches about 182k tokens, against 12–16k with search, and a run costs 8–9 times as much at the same pass rate (§6.6). The second, which harnesses ship, is search: definitions stay out of the prompt and the agent finds tools on demand (Table 1). Most shipped search is BM25 [2, 20, 22]. BM25 is cheap and needs no model, but it is imprecise, so agents search repeatedly, and each search is a full LLM turn that re-reads the whole context. With a frontier agent model, a wasted search costs far more than the search itself.

Decision models, or DMs, read a state and a question with named options and return a probability per option in one forward pass, without generating text. They are cheap and fast, and choosing among named options is what tool selection asks for. Our thesis is that a DM can serve as a drop-in, harness-agnostic tool selector: one API call per search, no training, and no change to the agent. We test it with Microsoft Decision 1 [16].

Table 1: Tool-selection mechanisms and the conditions that reproduce them. Used in: M, main tests in Hermes Agent, §6.3–§6.4; R, offline retrieval, §6.2; O, opencode, §6.6; E, the earlier six-ranker run, App. E.

Condition	Shipped or proposed in	What the agent sees and does	Used in
native	Cursor, Cline, opencode; Claude Code with search off	every tool definition in every request	O
cc	Claude Code ToolSearch [3]	names of all deferred tools; selection by exact name or keyword search weighted towards names; our approximation, since the exact scoring is undocumented	R O E
bm25	Anthropic tool search [2], Codex CLI [22], Hermes Agent [20]	Okapi BM25 [26] over names, descriptions and parameters	M R O E
groups	GitHub Copilot virtual tools [15]	one activation tool per server group; activating loads the group's definitions	O
embed	RAG-MCP [11]; Goose in 2025, since removed [6]	cosine similarity of query and tool embeddings	R O E
llm	Goose in 2025, since removed [6]; AnyTool-style routing [9]	an LLM reads one line per tool and names the best ones	R O
dm	this paper, first version	Microsoft Decision 1 ranks servers, then tools	R O E
selector F	this paper	Microsoft Decision 1 chooses services with a "none" option, then tools; five tools with descriptions plus names, §3	M

Contributions.

- A DM tool selector, selector F, and the analyses behind its design: choose services first with a "none" option, then ask two questions per service, and fill a fixed-size answer with usage-based names (§3, §6.1).
- A test suite built on the real tool definitions of 30 MCP servers, with simulated back ends, 38 graded coding tasks and isolated agent runs (§4).
- A protocol for measuring token differences between agent configurations: same-time pairs, task-level analysis, power planning and written analysis plans with logged interim looks (§5).
- Results for two agent models in Hermes Agent: selector F used 12.6% fewer tokens than BM25 with DeepSeek V4.1 Flash and 9.7% fewer with GPT-6.1 Sol, at equal pass rates (§6.3, §6.4).

The scope is limited: coding agents, simulated services, two agent models and one harness for the main tests (§8). "Universal" names the proposal, a selector that needs nothing from the harness but a query, not a result we have shown for every setting.

2 Background: how harnesses select tools, and decision models

Table 1 lists the mechanisms harnesses ship and the conditions we use to reproduce them. All but native loading give the agent a search tool and a way to call what it finds, usually with a short overview of the connected servers, so the agent pays for a definition only when it retrieves it.

Hermes Agent's bridge. Hermes Agent 0.19.0 defers MCP tools once their definitions exceed 10% of the context window, as hundreds of tools do. Its `tool_search` returns compact hits, a name and one line each; `tool_describe` returns one full definition; found tools are then called directly with typed arguments. Only the ranker behind `tool_search` differs between our main conditions.

Decision models. A request to OpenRouter's Decisions API [24] carries a *state*, here the agent's search query, and one or more named questions. A multiple-choice question lists up to 255 options, each with a free-text description, and returns a probability per option; a yes/no question returns one probability. Microsoft Decision 1 is post-trained from Qwen3.5-9B [16] and costs about $0.04 per million input tokens. We also benchmarked TypeSafe Jev [29] offline at 1,076 tools: both put the right tool first for 87.5% of queries and in the top five for 95.8%; Jev answered twice as fast, a median of 0.27 against 0.57 s, but cost $0.211 against $0.145 per 1,000 searches and wrongly answered "none" for 1.7% of queries against 0%. We use Microsoft Decision 1 throughout.

3 The DM tool selector

Selector F answers each `tool_search` query in two rounds of DM calls (Figure 1).

Service step. One multiple-choice question, "Which service has the tool the agent needs?", lists a card for each of the 30 services and a "none" option. If $P(\text{none}) \geq 0.4$ the answer is empty. Otherwise the selector takes services in order of probability until they cover 90% of the probability, at most three, and always includes a service that the query names. Card quality matters: the original cards confused GitHub, Git and GitLab, and the fixed cards say that GitHub hosts branches, commits and files, that Git is only the local working copy, and that GitLab hosts legacy projects (§6.5).

Tool step. For each chosen service, in parallel, two multiple-choice questions over its tools, each tool described by its name, the start of its description and its parameter names: "Which tool best serves what the agent wants to do?" and "Which tool would an expert engineer call first to do what the agent wants?". Scores are weighted by the service probability. The answer holds the five tools that best serve the query, with Hermes's one-line descriptions, and then, as names only, up to three prerequisites from the second question and the chosen services' most-used tools, at most 15 tools in all. Tools already shown in the run are not repeated. The usage lists are built leave-one-task-out from an earlier run, so a task never sees usage from its own runs.

Why this shape. Agents work service by service and use three to four tools per service, most of them in later steps; descriptions are expensive and names are cheap; and the DM's probabilities are too peaked to size a list, so the list size is fixed and filled by prerequisites and usage. The offline study of §6.1 chose each element.

Cost. A search takes about three DM calls in two sequential rounds. In the main tests a search took a median of 1.0 s and cost $0.36 per 1,000 searches; 5.7% of searches got the empty "none" answer.

4 Experimental setup

Tool catalog. We started each MCP server as an unprivileged user with dummy credentials and saved its `tools/list` response: names, descriptions and input schemas exactly as published. No tool was ever executed against a real service. The main catalog has 613 tools from 12 popular servers; 18 further servers bring it to 1,076 tools, including decoys such as GitLab beside GitHub, AWS CloudWatch beside Grafana and AWS EKS beside Kubernetes. App. B lists all servers.

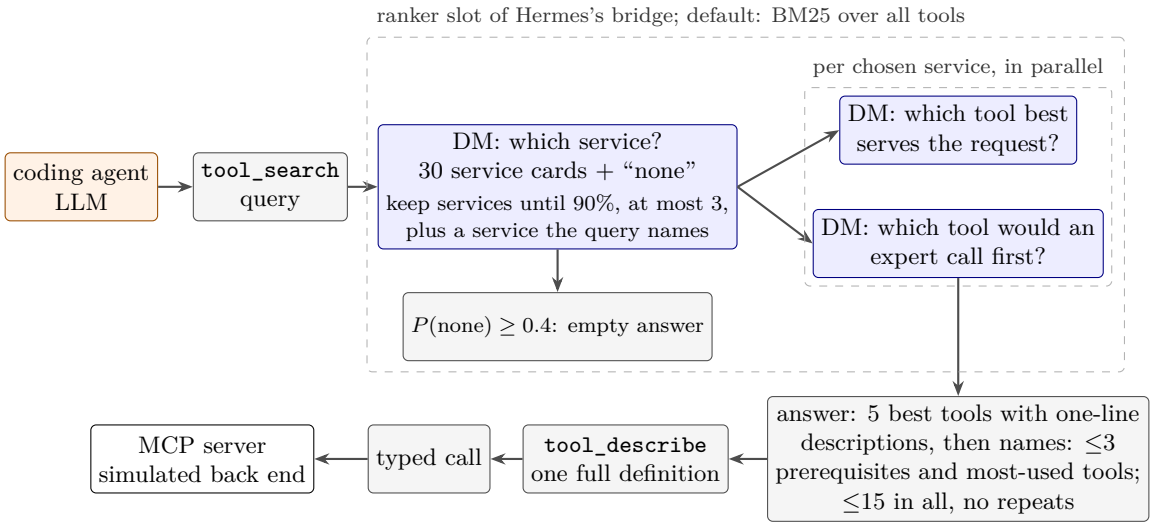


Figure 1: Selector F inside Hermes Agent's search bridge. The agent's query goes to the ranker slot. One DM call chooses the services or answers "none"; then two DM calls per chosen service run in parallel. The agent receives five compact hits and a list of names, reads full definitions with `tool_describe` and calls tools directly. In the baseline, BM25 fills the same slot; everything outside the slot is identical.

Simulated back ends. Each task has a world of rules that map a tool and matching argument values to a response, followed by organisation-wide facts shared by all tasks. Unmatched reads return an empty list or "not found"; unmatched writes succeed and are recorded for grading. Arguments are validated against the tool's schema.

Tasks. Each task is a small Python repository whose fix needs information that exists only in the services, such as a failing CI check, a Sentry stack trace or a decision in Slack. Task set v2 has 38 *symptom-first* tasks that do not name the service: 23 single-service, 7 multi-service, 4 *absent*, where the information does not exist and the right outcome is to stop and leave a `TODO`, and 4 controls that need no catalog tool (App. B). Every repository carries the same short handbook of where things live, like an `AGENTS.md`, and every prompt starts with the same note that the company's services are connected as tools, some loaded on demand through tool search, and ends with the same escape clause: if the information is not available, do not guess. One task, `k8s_crashloop`, is excluded from the main tests for both conditions, because its grading rule forbids a reasonable read-only `kubectl exec`. The offline queries and the opencode runs use the earlier set v1, 33 tasks whose prompts often name the service.

Grading. A run passes when the hidden tests pass, every required write was made with the right arguments, and no destructive call such as a merge, delete or rollout restart was made. We checked that each task's hidden tests fail on the original repository and pass with the reference solution, and that the reference calls get the intended answers.

Harnesses and isolation. The main tests use Hermes Agent [20] 0.19.0 with its own tool-search bridge over all 1,076 tools; we patched its search function to ask our server's ranking endpoint, so only the ranker differs. Supporting runs use opencode [23] 1.19.4. Agents run as an unprivileged user without credentials, each Hermes run in its own mount namespace, with a per-run token that the agent's shell cannot read and a deny list of probing commands (App. A).

Agent models and baseline. DeepSeek V4.1 Flash [8] is pinned to DeepInfra's fp8 endpoint with temperature 0.6; GPT-6.1 Sol is pinned to OpenAI with default sampling. The baseline is Hermes Agent's built-in BM25, unmodified. *Tokens* are all uncached input, cached input and output tokens summed over a run's LLM calls; *cost* is the LLM cost at the pinned provider's prices, including cache writes where billed, plus the DM fee.

5 Measuring token differences in agents

Pass rates are near ceiling for both rankers, so the differences we look for are in tokens, and tokens are noisy.

Run-to-run noise. Two runs of the same task under the same condition can differ by a large factor. The standard deviation of the per-slot log ratio between conditions was 0.55 with DeepSeek, so a single slot easily swings by a factor of 1.7, and 0.29 with Sol (Figure 3).

Drift. Batches run at different times are not directly comparable. In an A/A comparison, BM25 against BM25, the runs of the recommendations batch of App. D used 0.94× the tokens of BM25 runs made for the first version of this paper some days earlier, [0.87, 1.02] over 111 matched slots of 37 tasks, $p = 0.16$. That is not significant, but the interval admits differences as large as the effects we measure. Two BM25 batches run a day apart agreed closely, 1.01× [0.97, 1.06] over 555 slots.

Protocol. Both conditions run the same task and repeat number back to back, a *slot*. For each slot we take $\log(\text{tokens}_F/\text{tokens}_{\text{BM25}})$, average it per task, and test the 37 task means with a two-sided one-sample t-test; the 95% interval is a task-level bootstrap with 10,000 resamples, reported as a geometric-mean ratio. A verdict of "fewer tokens" needs $p < 0.05$ and an interval that excludes 1. Treating slots as independent would overstate the evidence, since repeats of a task are correlated. Before each batch we wrote an analysis plan, logged every interim look, and ran the batch to completion regardless of interim numbers (App. C). Runs that ended in provider errors without recorded tokens were rerun; nothing else was excluded.

Power. Simulations from the measured variance planned 15 repeats for DeepSeek, 98% power for a 12.5% token reduction, and 5 for Sol, about 98% at Sol's noise but only 80% at DeepSeek's.

6 Results

6.1 Offline: designing the selector

We compared 13 selector variants on three fixed query sets at 1,076 tools: 120 hand-written queries for the steps of task set v1, plus 40 that no tool can serve; 400 searches made by agents in the BM25 runs of App. D, labelled with the new tools the agent used after each search and before its next one; and 350 agent queries that name exactly one service, 200 of them Git, GitHub or GitLab. Table 2 gives the results.

Cards and the service nucleus matter most. Taking services until 90% of the probability instead of only the top one raised hit@5 from 0.90 to 0.96, and fixing the GitHub, Git and GitLab cards raised hit@1 from 0.88 to 0.92. A rule that always includes a named service put it first on all 350 named queries.

Table 2: Offline selector study at 1,076 tools with Microsoft Decision 1. Hand: hit@k on 120 hand-written queries. Agent: share of the next tools of 400 agent searches covered by the first 5 or all 15 tools, and share of searches with all next tools covered. Named: the named service ranked first, on 350 queries and on the 200 in the Git family. Cost: DM fee per 1,000 searches. Variants 4–13 build on variant 3. Bold: best per column.

	Hand			Agent			Named		
Variant	hit@1	hit@5	hit@15	first 5	all 15	all covered	all	Git family	$/1k
1 top service only	0.87	0.90	0.91	0.80	0.83	0.75	0.97	0.95	0.18
2 services until 90%	0.88	0.96	0.96	0.83	0.85	0.78	0.96	0.94	0.20
3 2 + fixed cards	**0.92**	0.98	0.99	0.83	0.86	0.79	0.98	0.97	0.19
4 “choose the named service”	**0.92**	0.98	0.99	0.83	0.86	0.79	0.98	0.97	0.20
5 “none” option	**0.92**	0.98	0.99	0.83	0.86	0.78	0.98	0.97	0.19
6 named-service rule	**0.92**	0.98	0.99	0.83	0.86	0.79	**1.00**	**1.00**	0.19
7 “expert first” question	0.83	**0.99**	**1.00**	0.83	0.85	0.78	0.98	0.98	0.20
8 yes/no per tool	0.75	0.97	**1.00**	0.82	0.86	0.80	0.98	0.97	0.29
9 sliding window, no service step	0.86	**0.99**	**1.00**	**0.84**	0.86	**0.81**	0.98	0.99	2.26
10 yes/no on all tools, no service step	0.63	0.86	0.95	0.73	0.83	0.76	0.67	0.67	3.43
11 “which tools are relevant”	0.88	0.98	0.99	0.83	0.86	0.79	0.98	0.97	0.20
12 best serves and expert first	**0.92**	0.98	**1.00**	0.83	0.86	0.79	0.98	0.97	0.36
13 best serves and relevant	**0.92**	0.98	0.99	0.83	0.86	0.79	0.98	0.97	0.36

The “none” option is free. It answered “none” for 95% of the 40 queries that no tool serves, never for a query that one does, and left every other column unchanged.

Wording matters little; other shapes are worse or costlier. Rephrasing the questions moved hit@1 by a few points at most. Asking yes or no per tool lowered hit@1 to 0.75 within services and to 0.63 over all tools, where it also lost the named service a third of the time. A sliding window over all tools, without a service step, covered slightly more next tools but cost about 12 times as much as variant 3, at about nine DM calls per search; yes/no over all tools needed five calls and cost about 18 times as much. Service-based variants needed about 2.1–2.2 calls per search.

Selector F combines the winners and was checked on held-out searches. Selector F is variant 3 with the “none” option of variant 5, the named-service rule of variant 6, the “expert first” question for prerequisites as names, and usage lists; its first five tools come from the same question as variants 3–6, so its hit@1 on hand-written queries is 0.92. On 500 held-out agent searches from the runs of the first version of this paper, selector F covered the most next tools within 15, 0.845, with all next tools covered for 79% of searches, against 0.806 and 75% for BM25; within the first five it covered 0.79 against 0.71, and it ranked the named service first for 99.7% of named queries against 85% for BM25 (Figure 2). It needs about three DM calls per search.

6.2 Offline: DM against shipped mechanisms and an LLM router

On the 120 hand-written queries, a server-then-tool DM hierarchy, the first version of our selector, ranked better than every shipped mechanism and an LLM router that reads one line per tool (Table 3). At 613 tools it put the right tool first for 93% of queries, against 48–54% for the Claude Code-style, BM25 and embedding mechanisms and 86% for the router, which cost three times as much per query and took 2.4 times as long. Adding 463 decoy-rich tools lowered hit@1 by 3–6 points for every mechanism and left the order unchanged. General-purpose embeddings did poorly, as ToolRet found [27]. On 183 queries written by agents in live runs the gap was smaller: hit@1 0.60

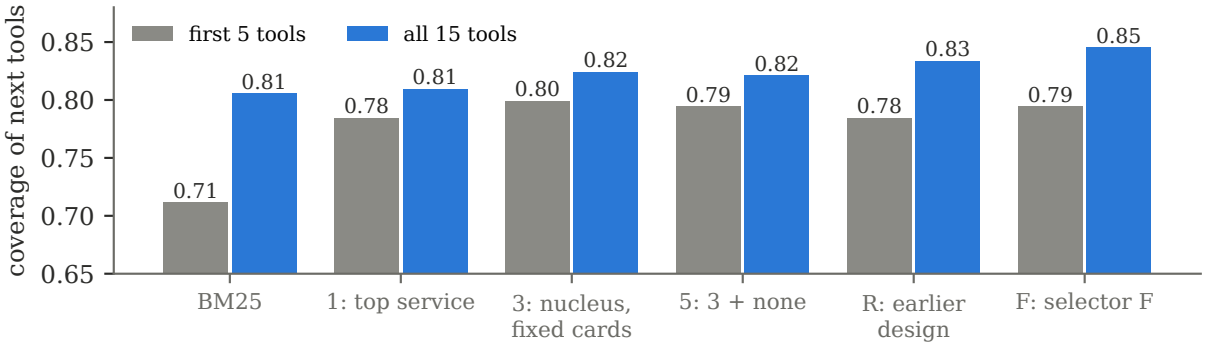


Figure 2: Held-out check on 500 agent searches from a different batch: share of the next tools covered by the first 5 and by all 15 returned tools. Variant numbers as in Table 2; R is the earlier recommendations design of App. D. The vertical axis starts at 0.65.

Table 3: Offline retrieval on 120 hand-written queries. Hit@k: a right tool is among the first k. Cost per 1,000 queries and median latency of the selecting call at 613 tools; name, BM25 and embedding search run locally. Bold: best per accuracy column.

	613 tools		1,076 tools			
Mechanism	hit@1	hit@5	hit@1	hit@5	$/1k queries	latency, s
Claude Code style	0.48	0.71	0.45	0.65	0	0.01
BM25	0.54	0.75	0.49	0.76	0	<0.01
embeddings	0.51	0.80	0.47	0.75	≈0	0.67
LLM router	0.86	**0.98**	0.82	**0.98**	0.44	1.31
DM hierarchy	**0.93**	0.97	**0.88**	0.97	0.15	0.54

for the DM against 0.45 for BM25, and hit@5 0.70 against 0.66, because agents write keyword-style queries that suit lexical search.

6.3 Main test: DeepSeek V4.1 Flash, 15 repeats

Each of the 37 tasks ran 15 times under BM25 and under selector F, 555 slots and 1,110 runs. Selector F used fewer tokens: 0.874× [0.82, 0.93], $p = 0.0002$, a 12.6% reduction, with fewer tokens on 361 of 555 slots (Table 4, Figure 3). Agents made 3.9 fewer searches per run, 7.1 against 11.1, made 8% fewer LLM calls and sent 5% smaller prompts per call. They read 1.3 more full definitions per run with `tool_describe`, which cost far less than the searches saved. Pass rates were equal, 550 of 555 each, with 5 discordant slots each way. Wall time did not change: the DM's second per search was offset by fewer turns. Cost per run rose by 20%, 1.20× [1.13, 1.28], because the DM fee of $0.0025 per run is large next to an agent that costs under a cent per run; §6.7 discusses this. The saving was largest on the 7 multi-service tasks, 0.76×, against 0.88× on the 22 single-service tasks.

6.4 Second agent model: GPT-6.1 Sol, 5 repeats

The same design with GPT-6.1 Sol and 5 repeats, 185 slots, also showed fewer tokens: 0.903× [0.84, 0.96], $p = 0.004$, a 9.7% reduction (Table 5). Sol searched less than DeepSeek to begin with, 5.5 times per run with BM25, and 2.2 times with selector F, 60% fewer. It made 12% fewer LLM calls, its prompts per call did not change, and pass rates were equal, 172 and 173 of 185. Cost did not

Table 4: Main test with DeepSeek V4.1 Flash: 37 tasks, 15 repeats, 555 same-time pairs. Per-run metrics as mean / median. Ratio: geometric mean of per-slot ratios of selector F to BM25, averaged per task, with a task-level bootstrap 95% CI; p: t-test over the 37 task means; for searches, the paired difference per run. Cost includes the DM fee. Bold: the better value where the difference is significant.

Metric	BM25	selector F	selector F vs BM25 [95% CI]	p
Passed runs	550/555	550/555	5 vs 5 discordant	1.00
Tokens per run	791k / 553k	**680k / 507k**	0.874× [0.82, 0.93]	0.0002
LLM calls per run	29.0 / 26	**26.8 / 24**	0.92× [0.88, 0.96]	0.0007
Searches per run	11.1 / 10	**7.1 / 6**	−3.9 [−5.0, −3.0]	<0.0001
Prompt tokens per call, median	21.1k	**20.1k**	0.95× [0.93, 0.98]	0.0006
Cost per run, $	**0.0082 / 0.0068**	0.0100 / 0.0089	1.20× [1.13, 1.28]	<0.0001
Wall time per run, s	137 / 120	140 / 120	1.01× [0.96, 1.07]	0.75

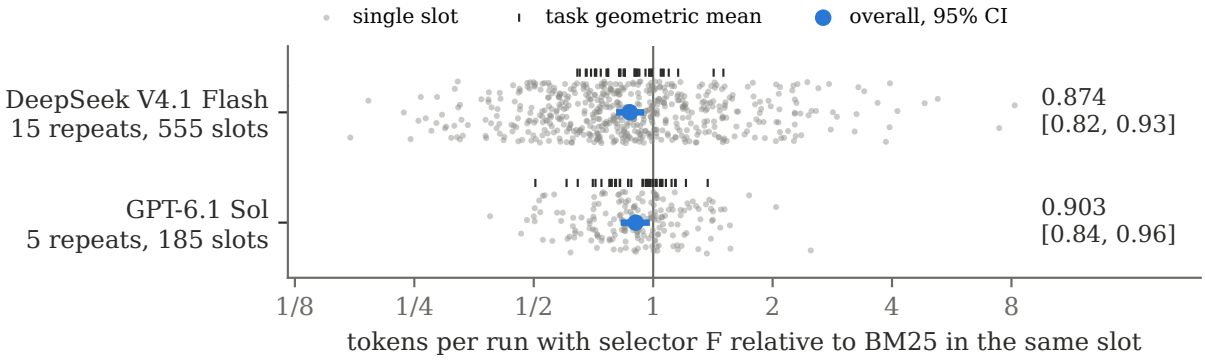


Figure 3: Tokens per run with selector F relative to BM25 in the same slot. Grey: single slots; black ticks: the 37 task-level geometric means; blue: overall geometric mean of the task means with a task-level bootstrap 95% CI. With DeepSeek, single slots range from 0.12 to 8.2, so only paired averages over many slots are informative.

change, 0.99× [0.95, 1.03], since Sol's fee per run was small next to its LLM cost and since cache writes, which selector F does not reduce, made up more than half of that cost.

For comparison at equal size, the plan fixed in advance a random draw of 5 of the 15 DeepSeek repeats per task. On that subsample the reduction was 0.915× [0.84, 1.01], $p = 0.07$: similar in size but not significant, as the power estimate for DeepSeek's higher noise predicted. The full 15-repeat result remains DeepSeek's verdict.

6.5 Failure analysis: the Git family

An earlier design, which took only the top service and lacked the fixed cards, failed `gh_open_pr` in 3 of 15 runs while BM25 passed all 15 (App. D). We asked the DM's service question for every distinct query from that batch that names exactly one service, 2,477 queries. The DM chose the named service for 97.9%. Of its 51 errors, 47 were within Git, GitHub and GitLab, most often Git for a GitHub query, and in 45 the named service came second. By service it chose correctly 608 of 639 times for GitHub, 86 of 95 for Git and 325 of 332 for GitLab, and in every case for Atlassian, Sentry, Slack, Notion, CloudWatch, MongoDB, Postgres and EKS. The errors came from the service cards, which did not say that pull requests and branches live on GitHub rather than in the local Git copy. With the fixed cards and the named-service rule, selector F passed `gh_open_pr`

Table 5: GPT-6.1 Sol, 37 tasks, 5 repeats, 185 same-time pairs, next to the pre-specified DeepSeek subsample of 5 of 15 repeats per task, also 185 pairs. Per-run metrics as mean / median; ratios and p as in Table 4. Cost includes cache writes and the DM fee. Bold: the better value where the difference is significant.

	GPT-6.1 Sol				DeepSeek, 5 of 15 repeats	
Metric	BM25	selector F	selector F vs BM25	p	selector F vs BM25	p
Passed runs	172/185	173/185	4 vs 5 discordant	1.00	184 vs 182 passed	0.63
Tokens per run	193k / 168k	**171k / 133k**	0.903× [0.84, 0.96]	0.004	0.915× [0.84, 1.01]	0.07
LLM calls per run	13.4 / 13	**11.5 / 10**	0.88× [0.83, 0.92]	<0.0001	0.95× [0.89, 1.01]	0.07
Searches per run	5.5 / 5	**2.2 / 2**	−3.3 [−4.5, −2.2]	<0.0001	−3.4 [−4.5, −2.4]	<0.0001
Cost per run, $	0.0724 / 0.0641	0.0715 / 0.0630	0.99× [0.95, 1.03]	0.63	1.25× [1.15, 1.36]	<0.0001
Wall time per run, s	53 / 45	52 / 41	0.99× [0.92, 1.05]	0.69	1.04× [0.95, 1.14]	0.38

Table 6: opencode on task set v1, 613 tools, two repeats per task. Pass: runs passed on the 25 tasks that need a catalog tool. Largest prompt: median over those runs. Cost: mean LLM cost per run over all 33 tasks, without the selector's fee. Bold: smallest largest prompt and lowest cost among conditions with catalog access.

	DeepSeek V4.1 Flash			Tencent hy4		
Condition	Pass	Largest prompt	$/run	Pass	Largest prompt	$/run
no catalog	0/25	9.6k	0.0110	0/25	15.4k	0.0154
native	25/25	181.9k	0.0722	25/25	188.5k	0.1039
cc	25/25	16.2k	0.0089	25/25	**32.0k**	**0.0375**
bm25	25/25	13.4k	0.0093	25/25	33.6k	0.0417
groups	25/25	29.0k	0.0179	24/25	52.7k	0.0562
embed	25/25	13.3k	0.0090	23/25	49.0k	0.0514
llm	25/25	**11.7k**	**0.0076**	25/25	32.6k	0.0419
dm	25/25	14.0k	0.0082	25/25	34.6k	0.0470

in 15 of 15 runs. The remaining failures are hard for both rankers: with Sol, `loki_errors` and `prom_threshold` account for 8 of BM25's 13 failures and 10 of selector F's 12; with DeepSeek, `prom_threshold` failed 3 times under selector F and 2 times under BM25.

6.6 Loading all tools: opencode

Supporting runs with opencode on the 33 v1 tasks and 613 tools, two repeats per task and two inexpensive agent models, show why harnesses search (Table 6). Without a catalog no tool task passed; with one, every condition passed 96–100% of runs. Loading every definition made the largest prompt about 182k tokens with DeepSeek; with ranked search it was 12–16k, and the cost per run fell from $0.072 to $0.008–0.009, 8–9 times less. Pass rate cannot separate the search mechanisms, which is why the main tests compare tokens.

6.7 Cost by agent model

The DM fee is fixed per search; the saving scales with the agent model's price. Table 7 gives the two measured results and prices the mean DeepSeek token mix at other models' list prices. With DeepSeek, selector F cut the LLM cost by 8.5% before the fee, less than the token reduction because the tokens saved are mostly cached input, which is cheap; the fee of $0.0025 per run then made the mean run 22% more expensive, or 20% as the paired ratio of Table 4. With Sol, the fee was $0.0008 per run and cost was unchanged. If a model produced DeepSeek's token mix, selector F

Table 7: LLM cost per run with BM25 and with selector F, DM fee included. Measured: the main tests, mean per run. Projected: the mean DeepSeek token mix per run, 27.5k uncached input, 755.9k cached input and 11.1k output tokens with BM25 and 24.8k, 648.4k and 10.8k with selector F, priced at list prices in $ per million tokens, plus the measured fee of $0.0025 per run. [a]Also includes cache writes at $2.50 per million tokens.

	Agent model	Price, in / cached / out	BM25 $/run	selector F $/run	Change
Measured	DeepSeek V4.1 Flash, DeepInfra	0.098 / 0.003 / 0.29	**0.0082**	0.0100	+22%
	GPT-6.1 Sol[a]	2.00 / 0.10 / 10.00	0.0724	0.0715	−1%
Projected	DeepSeek V4.1 Flash, list price	0.15 / 0.003 / 0.60	**0.0131**	0.0147	+12%
	Claude Haiku 5.5	0.10 / 0.01 / 0.50	**0.0159**	0.0169	+6%
	Claude Sonnet 5.5, GPT-6.1 Sol	2.00 / 0.10 / 10.00	0.2420	**0.2247**	−7%
	Kimi K3.1 Pro	2.00 / 0.20 / 10.00	0.3176	**0.2895**	−9%
	Claude Fable 5.1	10.00 / 0.25 / 50.00	1.0211	**0.9510**	−7%

would make runs 6–12% more expensive with inexpensive models and 7–9% cheaper with mid-range and frontier ones, saving $17–70 per 1,000 runs. Sol's measured result is less favourable than this projection because its cost was dominated by cache writes, so the projection is indicative only.

7 Discussion

Fewer searches matter more than cheaper searches. A search costs the agent a full turn: the model re-reads its whole context to write the query and again to read the answer. Selector F cut searches by 36% with DeepSeek and 60% with Sol; the tokens saved came from fewer LLM calls, not from shorter answers, since selector F's answers are longer than BM25's. A better first answer, with the right service's likely next tools listed by name, saves the turns that a rephrased query would cost.

When the DM pays off. The fee is fixed per search, while the saving is a share of the agent's own cost. The DM therefore pays off with expensive agent models, many tools and search-heavy work, and does not with inexpensive models such as DeepSeek V4.1 Flash, whose whole run costs about a cent. An efficient agent that searches little, like Sol, gains fewer searches to remove, and where cache writes dominate cost, as with Sol, fewer cached reads save little.

What "universal" means here. Selector F needs one API call per question, no training, and only the query and the catalog; it worked unchanged with two agent models and slots into any harness that has a search tool. We have not shown that it helps in every harness, catalog or kind of task.

What the DM is good and bad at. The DM is good at comparing options: choosing a service among cards or a tool among a service's tools. It is poor at judging a tool in isolation, as the yes/no variants showed, and at predicting the next step of an agent, where always choosing the shell did better (App. F). Its answers depend on the option texts: the Git-family errors came from service cards, not from the model.

8 Threats to validity

Scope. The services are simulated: real services return longer and messier results, which would raise token counts and could change how agents search. The tasks are coding tasks only. Two agent models were tested, and the main tests used one harness, Hermes Agent; opencode results come from an earlier selector and task set.

Design choices made before the tests. Selector F was designed in exploratory runs on a separate long task and in the offline study, before the main tests; we checked it on held-out searches from a different batch. The usage lists come from the same task suite but are built leave-one-task-out. The connected-services note in every prompt may raise search rates above those of an unprompted agent; it is the same for both conditions.

Exclusions and interruptions. `k8s_crashloop` was excluded for both conditions, as stated in each plan. Interim looks, container restarts and reruns after rate-limit errors are logged in the plans (App. C); no run was dropped for its result.

Drift and baselines. Our A/A comparison could not rule out drift of the size we measure, so only same-time pairs support the claims; the earlier six-ranker comparison of App. E does not meet this standard. The baseline is Hermes Agent's BM25 as shipped; a tuned BM25 or query rewriting might narrow the gap.

9 Related work

Tool retrieval for large catalogs. Retrieving tools before generation is the standard answer to large catalogs [7, 25]. AnyTool [9] selects among over 16,000 APIs with a hierarchy of LLM agents, category first and tool second, and MCP-Zero [10] brings the server-then-tool idea to MCP; selector F applies the same hierarchy with a decision model instead of an LLM. RAG-MCP [11] retrieves MCP tool descriptions with embeddings to reduce prompt bloat, and Mudunuri et al. [19] select among 121 tools of five MCP servers with embeddings, without agent runs. ToolRet finds that general-purpose embedding retrievers do poorly on tool retrieval [27], and Lee [13] find that selecting a tool is largely retrieval. Agent skills [1] apply progressive disclosure to instructions; our service cards and handbook play this role for the catalog.

MCP benchmarks. LiveMCPBench [17], MCPMark [30], MCP-Atlas [5] and Toolathlon [14] score agents' task completion with real MCP tools. We fix the agent, vary only the selector, and simulate back ends so that runs are reproducible and writes can be graded safely.

Vendor reports. Stacklok compared its MCP Optimizer with Anthropic's tool search on 2,792 tools [28], and Arcade tested Anthropic's tool search on about 4,000 tools [12]. Both measure selection accuracy, not task completion or tokens per task.

Decision models. The Artificial Analysis Decision Index [4] and an analysis of it [21] compare DMs with LLMs on standalone decisions, including tool calling. We place a DM inside agents as the tool selector and measure its effect on whole runs.

10 Conclusion

A decision model can serve as an agent's tool selector. Built on Microsoft Decision 1, selector F chooses services with a "none" option, then asks two questions per service and adds usage-based names, at about three DM calls per search. Offline it put the right tool first for 92% of hand-written queries at 1,076 tools, against 49% for BM25. Inside Hermes Agent, in pre-planned same-time paired tests, it used 12.6% fewer tokens than BM25 with DeepSeek V4.1 Flash and 9.7% fewer with GPT-6.1 Sol, with 36–60% fewer searches and equal pass rates. Its fee made runs 20% more expensive with the inexpensive model and left cost unchanged with the frontier model; the saving grows with the agent model's price and with how much it searches. These results hold for coding agents with simulated services, two agent models and one harness; testing other harnesses, real services and non-coding work is the next step.

Reproducibility. We will release the tool harvester, catalog snapshots, simulator, tasks with hidden tests and reference solutions, the selector and ranking server, the Hermes patch, the sandbox setup, all run records, the analysis plans and the analysis and figure scripts.

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A Sandbox hardening

The isolation of §4 is the result of fixes made during development; runs made before them are not reported. In order:

- Agents run as an unprivileged user with an empty environment and no credentials; experiment files are readable only by root and the API key is held by a local proxy that admits only the experiment models, because harness-level permissions alone let an agent read hidden tests through absolute paths.
- Each Hermes run gets its own mount namespace with a private work directory, home directory and `/tmp`, so runs cannot see each other's files or leftovers.
- The MCP server accepts only requests that carry a per-run token the agent's shell cannot read, so the shell cannot list the catalog or query the ranker directly.
- Hermes's deny list blocks probing commands: `curl`, `wget`, `ps`, `ss`, `netstat`, `nmap`, `lsof`, `/proc`, `localhost` and `find /`.
- Rankers are compared inside Hermes's own bridge rather than through a separate search tool with a generic call tool, so that only the ranking differs.
- The provider and sampling are pinned as described in §4.

B Tool catalog and task set

Table 8: The real tool catalog: tools per MCP server, from each server's own `tools/list`.

Catalog	Servers with their numbers of tools	Tools
main	Linear 198, Atlassian 98, GitHub 91, Grafana 81, Git 28, MongoDB 27, Playwright 25, Notion 24, Kubernetes 23, Sentry 9, Slack 8, Postgres 1	613
extra	GitLab 127, monday 84, Salesforce 74, Heroku 33, HubSpot 21, AWS CloudWatch 19, AWS EKS 16, Airtable 16, Filesystem 14, CircleCI 13, Browserbase 9, Memory 9, Netlify 9, Terraform 9, Elasticsearch 4, ClickHouse 3, Figma 2, Docker 1	+463
all	30 servers	1,076

Table 9: Task set v2, 38 tasks. Prompts are shortened to their first sentence and shown without the escape clause. Services: the servers of the needed tools, in order; $+w$: number of required writes; $\star$: new in v2.

Category	Task	First sentence of prompt	Services
control	`c_chunk`	chunk in batching/chunk.py drops the last partial chunk.	–
control	`c_dedupe`	dedupe in collections_util/dedupe.py loses the original order.	–
control	`c_duration`	parse_duration in timeutil/duration.py only understands hours.	–
control	`c_leap`	is_leap in calendar_util/leap.py is wrong for some years.	–
single	`confluence_limits`	Our public API rate limits changed.	atlassian
single	`eks_importer` $\star$	The data team's nightly batch-importer job has failed three nights in a row.	aws_eks
single	`gh_ci_fail`	Pull request #214 in acme/payments can't be merged because a check is red.	github $+1w$
single	`gh_codesearch`	SUPPORTED_CURRENCIES in shop/currencies.py has drifted from the company-wide canonical list, which lives in another repository.	github
single	`gh_dependabot`	Security says acme/payments has dependencies with known vulnerabilities.	github
single	`gh_issue`	Please implement acme/payments#301.	github
single	`gh_open_pr`	Pagination in acme/catalog is broken (see #57).	github $+2w$
single	`gh_release`	We are behind on the ledger API.	github
single	`gh_review`	There is reviewer feedback waiting on pull request #88 in acme/ledger.	github $+1w$
single	`grafana_incident`	We are in the middle of an incident about checkout timeouts and the responders agreed on a mitigation.	grafana $+1w$
single	`jira_impl`	Pick up PAY-412: implement it, then move it to review.	atlassian $+1w$
single	`k8s_crashloop`	ledger-worker keeps restarting in production (namespace ledger) since the last config change.	kubernetes
single	`k8s_oom`	report-builder (namespace reports) keeps dying in production.	kubernetes

Category	Task	First sentence of prompt	Services
single	`linear_bug`	Fix ENG-77 and move it to In Review.	linear +1w
single	`loki_errors`	Customers stopped getting shipping e-mails at around 06:00 UTC today; the notifier service is the suspect.	grafana
single	`mongo_shape`	The nightly export has started crashing on recent orders.	mongodb
single	`notion_runbook`	We are rotating the webhook signing keys.	notion
single	`oncall_contact`	Set ESCALATION_CONTACT in ops/escalation.py to the e-mail address of whoever is on call for the payments team right now.	grafana
single	`pg_migration`	Add migrations/0042_payments_merchant_idx.sql with an index named payments_merchant_created_idx for looking up a merchant's payments by creation time.	postgres
single	`prom_threshold`	Set SLOW_REQUEST_MS in alerts/config.py to twice the checkout service's current p99 request latency, in milliseconds.	grafana
single	`sentry_events`	Since this morning's deploy some clients get HTTP 500 from POST /refunds.	sentry
single	`sentry_fix`	The noisiest production error of payments-api right now needs to go.	sentry +1w
single	`slack_decision`	The payments team agreed on a new webhook retry policy in a chat discussion.	slack +1w
multi	`cw_reports` ⋆	The legacy reports API has been answering 502 since yesterday evening.	aws_cloudwatch, slack +1w
multi	`gl_pipeline` ⋆	The main-branch pipeline of billing-legacy is red.	gitlab +1w
multi	`linear_notion_pr` ⋆	Do ENG-91 the way our API guidelines say, and open a pull request for it from a branch named eng-91 against main in acme/public-api.	linear, notion, github +1w
multi	`multi_alert`	Something is paging for the checkout service.	grafana, slack +1w
multi	`multi_jira_conf`	Implement PAY-530 as its design document specifies, then leave a one-line summary on the ticket.	atlassian +1w
multi	`multi_sentry_gh`	A recent change to ledger-worker introduced a reconciliation bug that shows up as a production error.	sentry, github +1w
multi	`promo_sentry_jira` ⋆	Android users report they can't apply promo codes.	sentry, atlassian +1w
absent	`absent_jira`	Implement PAY-999 in payments/limits.py.	–
absent	`absent_release`	Pin SDK_VERSION in sdk/version.py to the newest release of acme/ghost-sdk (without the leading v).	–
absent	`absent_sentry`	The billing-cron error someone mentioned in standup needs fixing in billing/cron.py.	–
absent	`absent_slack`	Set CACHE_TTL_S in cache/config.py to the value the platform team agreed on.	–

C Analysis plans and interim-look logs

The three plans below are reproduced verbatim, including their interim-look logs, as they stood when each batch's final analysis was run. File and condition names refer to the run records: `hermes_default` is BM25, `hermes_dmfrankenbridge` is selector F and `hermes_dmrec510bridge` is the recommendations design of App. D. The first plan was written after an interim look, which it records; the other two were written before their first run.

C.1 Recommendations design with DeepSeek V4.1 Flash

```
# Analysis plan: DM recommendations 5+10 vs Hermes BM25 (written before the final analysis)

Written: 2026-11-07T01:52Z, with 254 of 1,110 planned runs finished. Interim looks so far: ratio and
p-values were viewed at 103 slots (0.89x, p about 0.02). From now on only progress counts are viewed until the end.

## Design (fixed)
- Hermes Agent 0.19.0, task set v2, 1,076 tools, schema validation, deny list, connected-services note.
- DeepSeek V4.1 Flash pinned to DeepInfra fp8, temperature 0.6, seed 7.
- 37 tasks (k8s_crashloop excluded for both setups before the final analysis: its grading rule forbids a reasonable
  read-only exec), 15 repeats, two setups per (task, repeat) slot queued back to back: 555 planned slots.
- Usage lists for the recommendations are leave-one-task-out (built from the paper's main run, excluding the task).

## Primary analysis (the verdict)
- Measure: total tokens per run (input + cached input + output over all LLM calls).
- Per slot: log(tokens_recommendations / tokens_BM25). Per task: mean over its slots (37 values).
- Primary test: one-sample t-test of the 37 task means against 0, two-sided, alpha 0.05.
- Interval: task-level bootstrap (resample tasks, 10,000 resamples, seed 1) of the geometric-mean ratio, 95%.
- Verdict: "fewer tokens" only if the test gives p < 0.05 AND the interval excludes 1; otherwise "not established".
- Only complete slots (both setups finished) count; no other exclusions after this point.

## Secondary (reported, not used for the verdict)
- Cost per run including the DM fee (paired ratio, same method); pass rate per setup; searches; LLM calls; wall time.
- Slot-level paired t-test and Wilcoxon (they overstate the evidence because repeats of a task are correlated).

## Interim looks log
- 2026-11-07T02:13Z: look 2 at 347 runs (user request). Commitment: the batch runs to all 555 slots regardless; the verdict uses
↪  only the final analysis.
- 2026-11-07T03:27Z: look 3 at 669 runs (user request). Commitment unchanged.
- 2026-11-07T05:11Z: container restart interrupted the batch at 1076 runs; resumed with the same command (finished runs kept,
↪  interrupted ones rerun). No design or analysis change.
- 2026-11-07T05:25Z: 2 runs (prom_threshold r14 and sentry_fix r14, DM recommendations) ended with HTTP 429 rate-limit errors from
↪  the provider and recorded no tokens; treated as unfinished like the restart-interrupted runs and rerun before any final
↪  analysis. Their BM25 partners are kept.
```

C.2 Selector F with DeepSeek V4.1 Flash

```
# Analysis plan: combined DM selector F vs Hermes BM25 (written before any run)

Written: 2026-11-08T02:21Z, before the first run of this batch.

## Selector F (fixed before the run; chosen in an offline selector study and checked on held-out queries)
Microsoft Decision 1. Fixed service cards (GitHub lists branches/commits/files; Git is the local copy only; GitLab is
legacy hosting); services until 90% of the probability, at most 3; a 'none' option (empty answer, no BM25 fallback, when
P(none) >= 0.4); a service the query names is always included; per search the 5 tools that best serve the query with
descriptions, plus as names up to 3 prerequisites ('which tool would an expert engineer call first') and the services'
most-used tools (leave-one-task-out lists from the paper's main run), at most 15 tools; tools already shown in the run are
not repeated. Code: harness/selectors_real.py dm_franken; condition hermes_dmfrankenbridge.

## Design
Same as results/final_rec510: Hermes Agent 0.19.0, task set v2, 1,076 tools, schema validation, deny list,
connected-services note; DeepSeek V4.1 Flash pinned to DeepInfra fp8, temperature 0.6, seed 7; 37 tasks (k8s_crashloop
excluded for both setups, as before); 15 repeats; BM25 and F queued back to back per (task, repeat) slot: 555 slots.
The batch runs to completion regardless of interim numbers. Runs that end in provider errors with no recorded tokens
are rerun, as before; nothing else is excluded.

## Primary analysis (the verdict)
- Measure: total tokens per run. Per slot log(tokens_F / tokens_BM25); per task the mean over its slots (37 values).
- Test: one-sample t-test of the 37 task means against 0, two-sided, alpha 0.05; task-level bootstrap 95% interval
  (10,000 resamples, seed 1). Verdict "fewer tokens" only if p < 0.05 and the interval excludes 1.

## Secondary (reported, not part of the verdict)
- Pass rate per setup and McNemar test on discordant slots; failures listed per task (in particular gh_open_pr).
```

```
- Cost per run including the DM fee; searches; LLM calls; wall time; slot-level tests.
- Comparison with the earlier recommendations design (results/final_rec510) is descriptive only (different batches).

## Interim looks log
- 2026-11-08T02:34Z: container restart interrupted the batch at 61 runs; resumed with the same command (finished runs kept,
↪  interrupted ones rerun). No design or analysis change.
- 2026-11-08T02:35Z: look 1 at 62 runs (user request). Commitment: the batch runs to all 555 slots regardless.
- 2026-11-08T02:45Z: look 2 at 101 runs (user request). Commitment unchanged.
- 2026-11-08T03:11Z: look 3 at 225 runs (user request). Commitment unchanged.
- 2026-11-08T04:56Z: look 4 at 661 runs (user request). Commitment unchanged.
- 2026-11-08T06:35Z: look 5 at 1082 runs (user request). Commitment unchanged.
```

C.3 Selector F with GPT-6.1 Sol

```
# Analysis plan: DM selector F vs Hermes BM25 with GPT-6.1 Sol (written before any run)

Written: 2026-11-08T07:22Z, before the first run of this batch. The earlier 2-task Sol check (results/sol_check)
used tasks chosen because they favour F; it is not part of this test.

## Design
Identical to results/final_franken except the agent model and the number of repeats: GPT-6.1 Sol
(openai/gpt-6.1-sol, pinned to provider OpenAI, default sampling), 37 tasks (k8s_crashloop excluded), 5 repeats, BM25 and
F queued back to back per (task, repeat) slot: 185 slots, 370 runs. Selector F unchanged (harness/selectors_real.py
dm_franken). Cost per run now includes cache-write tokens at $2.50/M (fix verified against the key's charge).
The batch runs to completion regardless of interim numbers; provider-error runs with no tokens are rerun.
Power (simulated from measured variance): about 98% for a 12.5% token reduction at Sol-like noise (SD 0.30), 80% at
DeepSeek-like noise (SD 0.55).

## Primary analysis (the verdict, for Sol)
Same as results/final_franken: per-task mean of log(tokens_F / tokens_BM25) over its slots (37 values), one-sample t-test,
two-sided, alpha 0.05, plus task-level bootstrap 95% interval (10,000 resamples, seed 1). "Fewer tokens" only if p < 0.05
and the interval excludes 1.

## Secondary
Cost per run including the DM fee and cache writes (paired ratio, same method); pass rate and McNemar; searches; LLM
calls; wall time.

## Comparable DeepSeek table (descriptive; DeepSeek's own verdict remains the full 15-repeat result)
From results/final_franken, for each of the 37 tasks draw 5 of the 15 repeats at random with
random.Random(20261108).sample(range(15), 5), drawn task by task in alphabetical task order; both setups of a drawn slot
are kept. The same metrics are computed on that subsample and shown next to Sol.

## Interim looks log
- 2026-11-08T07:36Z: look 1 at 103 runs (user request). Commitment: the batch runs to all 185 slots regardless.
- 2026-11-08T07:44Z: look 2 at 184 runs (user request). Commitment unchanged.
- 2026-11-08T07:59Z: look 3 at 321 runs (user request). Commitment unchanged.
```

D An earlier design: recommendations

Before selector F we tested a recommendations design under the same protocol: only the top service, with the original cards; its five best tools with one-line descriptions; and up to ten names from the DM's next-best tools and the service's most-used tools. With DeepSeek V4.1 Flash, 37 tasks and 15 repeats, 555 same-time pairs, it used 0.910× the tokens of BM25 [0.85, 0.97], $p = 0.009$, a second, independent confirmation of a token reduction with a DM selector. Searches fell from 11.5 to 7.4 per run and LLM calls by 6%, 0.94× [0.90, 0.98]. Cost per run rose by 8.5%, 1.085× [1.02, 1.16], and wall time did not change, 1.04× [0.97, 1.11]. It passed 549 runs against 552 for BM25, with 6 and 3 discordant slots, McNemar $p = 0.51$; three of its failures were `gh_open_pr`, which led to the analysis of §6.5. Its 400 BM25-run searches are the agent set of the offline study, and its BM25 runs the A/A comparison of §5.

E The six-ranker comparison of the first version

The first version of this paper compared six rankers inside Hermes Agent's bridge with DeepSeek V4.1 Flash on all 38 tasks, three repeats each, 114 runs per ranker (Table 10). Two caveats apply.

Three repeats give too few pairs for effects of 10%, and only BM25, DM and random ran back to back in each slot, while the other three ran later in separate batches, so their differences from BM25 are within the drift we later measured (§5). The robust finding is that ranking matters: random ranking used 1.57× the tokens of BM25. All rankers, random included, passed 96.5–100% of runs.

Table 10: Six rankers in Hermes Agent's bridge, first version: 38 tasks, three repeats, 114 runs per ranker. Tokens: median per run. Ratio: geometric mean of per-slot token ratios against BM25 with a task-level bootstrap 95% CI. LLM calls and searches: mean per run. DM with "none" is the DM ranker with a "none" option. Not same-time for the last three rows.

Ranker	pass@1	Tokens	Ratio [95% CI]	LLM calls	Searches
BM25	0.991	558k	–	29.0	11.4
DM	0.991	480k	0.93 [0.82, 1.04]	27.9	9.7
random	1.000	1,051k	1.57 [1.38, 1.79]	34.9	12.8
DM with "none"	0.991	459k	0.88 [0.78, 0.98]	27.4	10.1
embeddings	0.965	562k	0.91 [0.81, 1.03]	27.9	11.1
Claude Code style	0.991	576k	1.02 [0.89, 1.16]	23.2	6.0

F Next-tool prediction in a real session

This experiment motivates using the DM to select tools rather than to call them. We took the 395 tool decisions of one real coding session and asked each model to predict the next tool among the 38 available. Microsoft Decision 1 reached an accuracy of 0.49 and TypeSafe Jev [29] 0.60, both below always choosing the shell at 0.75, although the DMs' top five contained the right tool 94–96% of the time. In that session 51 always-loaded tool definitions took about 59k tokens, 93% of the first request and 16% of all context tokens read. Most tool choices in a coding session are easy and the LLM must write every call's arguments; the DM is useful where the choice is hard, among hundreds of rarely used catalog tools.